

Volume Vi Lattice And Order Theory Reference

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Lattice Order

0.1 Breadcrumb

Why Lattice Theory Gets Its Own Home

0.1.1 Ordered Sets

Definition 0.1.1 (Partial Order). A relation $\leq$ on a set S is a *partial order* if it satisfies, for all $a, b, c \in S$:

1. **Reflexivity:** $a \leq a$.
2. **Anti-symmetry:** $a \leq b \wedge b \leq a \implies a = b$.
3. **Transitivity:** $a \leq b \wedge b \leq c \implies a \leq c$.

The pair $(S, \leq)$ is called a *partially ordered set*, or *poset*.

Definition 0.1.2 (Strict Partial Order). A relation $<$ on a set S is a *strict partial order* if it satisfies, for all $a, b, c \in S$:

1. **Irreflexivity:** $\neg(a < a)$.
2. **Asymmetry:** $a < b \implies \neg(b < a)$.
3. **Transitivity:** $a < b \wedge b < c \implies a < c$.

Definition 0.1.3 (Total Order). A **partial order** $\leq$ on a set S is a *total order* (or *linear order*) if it additionally satisfies the *law of trichotomy*: for all $a, b \in S$, exactly one of

$$a < b, \quad a = b, \quad b < a$$

holds. The pair $(S, \leq)$ is called a *totally ordered set* or a *chain*.

0.1.2 Chains and Antichains

Definition 0.1.4 (Chain). Let $(S, \leq)$ be a partially ordered set. A subset $C \subseteq S$ is a *chain* if every pair of elements in C is comparable:

$$\forall x, y \in C, \quad x \leq y \vee y \leq x.$$

Equivalently, the restriction of $\leq$ to C is a **total order** on C .

Definition 0.1.5 (Antichain). Let $(S, \leq)$ be a partially ordered set. A subset $A \subseteq S$ is an *antichain* if no two distinct elements of A are comparable:

$$\forall x, y \in A, \quad x \neq y \implies \neg(x \leq y) \wedge \neg(y \leq x).$$

0.1.3 Partial and Total Maps

Definition 0.1.6 (Partial Map). Let A and B be sets. A *partial map* $f : A \rightarrow B$ is a relation $f \subseteq A \times B$ such that every element of A has at most one image:

$$\forall x \in A, \forall y_1, y_2 \in B, (x, y_1) \in f \wedge (x, y_2) \in f \implies y_1 = y_2.$$

Definition 0.1.7 (Total Map). A *partial map* $f : A \rightarrow B$ is a *total map* if it is defined on every element of A :

$$\forall x \in A, \exists y \in B, (x, y) \in f.$$

A total map is written $f : A \rightarrow B$ and is precisely a function in the usual sense.

0.1.4 Maps Between Ordered Sets

Definition 0.1.8 (Order-Preserving Map). Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \rightarrow Q$ is *order-preserving* (or *monotone*) if

$$\forall x, y \in P, x \leq_P y \implies \varphi(x) \leq_Q \varphi(y).$$

Definition 0.1.9 (Order-Embedding). Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets. A map $\varphi : P \rightarrow Q$ is an *order-embedding* if

$$\forall x, y \in P, x \leq_P y \iff \varphi(x) \leq_Q \varphi(y).$$

When φ is an order-embedding, one writes $\varphi : P \hookrightarrow Q$.

Definition 0.1.10 (Order-Isomorphism). An *order-embedding* $\varphi : P \rightarrow Q$ that is surjective is an *order-isomorphism*. When an order-isomorphism exists, P and Q are said to be *order-isomorphic*, written $P \cong Q$.

Lemma 0.1.11 (Every Order-Embedding Is Injective). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be partially ordered sets, and let $\varphi : P \rightarrow Q$ be an order-embedding. Then φ is injective. Go to proof.*

0.1.5 The Covering Relation

Definition 0.1.12 (Covering Relation). Let $(P, \leq)$ be a partially ordered set and let $x, y \in P$ with $x < y$. The element y *covers* x , written $x \prec y$, if there is no element of P strictly between them:

$$x \prec y := x < y \wedge \neg (\exists z \in P, x < z < y).$$

Lemma 0.1.13 (Characterizations of Order-Isomorphisms for Finite Posets). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets and let $\varphi : P \rightarrow Q$ be a bijection. The following are equivalent:*

1. φ is an order-isomorphism.
2. For all $x, y \in P$: $x <_P y \iff \varphi(x) <_Q \varphi(y)$.
3. For all $x, y \in P$: $x \prec_P y \iff \varphi(x) \prec_Q \varphi(y)$.

Go to proof.

Proposition 0.1.14 (Hasse Diagram Characterization for Finite Posets). *Let $(P, \leq_P)$ and $(Q, \leq_Q)$ be finite partially ordered sets. Then P and Q are order-isomorphic if and only if their Hasse diagrams are isomorphic as directed graphs — that is, identical up to relabeling of elements. Go to proof.*

0.1.6 Duality

Definition 0.1.15 (Dual Ordered Set). Let $(P, \leq)$ be a partially ordered set. The *dual* (or *opposite*) of P , denoted P^{op} , is the set P equipped with the reversed order:

$$x \leq_{P^{\text{op}}} y \iff y \leq_P x.$$

Proposition 0.1.16 (Duality Principle). *Let Φ be a statement about partially ordered sets, expressed purely in terms of $\leq$, that holds in every partially ordered set. Then the dual statement Φ^{op} , obtained by replacing every occurrence of $\leq$ with $\geq$, also holds in every partially ordered set. Go to proof.*

0.1.7 Extremal Elements

Definition 0.1.17 (Bottom Element). Let $(P, \leq)$ be a partially ordered set. An element $\perp \in P$ is a *bottom element* (or *least element*) of P if

$$\forall x \in P, \quad \perp \leq x.$$

If a bottom element exists it is unique; it is denoted $\perp_P$ or simply $\perp$.

Definition 0.1.18 (Top Element). Let $(P, \leq)$ be a partially ordered set. An element $\top \in P$ is a *top element* (or *greatest element*) of P if

$$\forall x \in P, \quad x \leq \top.$$

If a top element exists it is unique; it is denoted $\top_P$ or simply $\top$.

Definition 0.1.19 (Maximal Element). Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *maximal in Q* if no element of Q strictly exceeds it:

$$\forall x \in Q, \quad a \leq x \implies a = x.$$

The set of maximal elements of Q is denoted $\text{Max}(Q)$.

Definition 0.1.20 (Minimal Element). Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $a \in Q$ is *minimal in Q* if no element of Q is strictly below it:

$$\forall x \in Q, \quad x \leq a \implies x = a.$$

Definition 0.1.21 (Greatest Element). Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *greatest element* (or *maximum*) of Q if it dominates every element of Q :

$$\forall x \in Q, \quad x \leq m.$$

If it exists it is unique; it is denoted $\max Q$.

Definition 0.1.22 (Least Element). Let $(P, \leq)$ be a partially ordered set and let $Q \subseteq P$. An element $m \in Q$ is the *least element* (or *minimum*) of Q if it lies below every element of Q :

$$\forall x \in Q, \quad m \leq x.$$

If it exists it is unique; it is denoted $\min Q$.

Proposition 0.1.23 (Every Nonempty Finite Poset Has a Maximal Element). *Let $(P, \leq)$ be a finite partially ordered set. Then every nonempty subset $Q \subseteq P$ has at least one maximal element. Furthermore, for every $x \in P$ there exists $y \in \text{Max}(P)$ with $x \leq y$. Go to proof.*

0.1.8 Lifting and Co-Lifting

Definition 0.1.24 (Lifting). Let $(P, \leq)$ be a partially ordered set. The *lifting* of P is the poset $P_\perp := P \cup \{\perp\}$, where $\perp \notin P$, with the order:

$$\perp \leq x \quad \text{for all } x \in P_\perp, \quad x \leq_{P_\perp} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P_\perp$ is P with a new **bottom element** $\perp$ adjoined.

Definition 0.1.25 (Co-Lifting). Let $(P, \leq)$ be a partially ordered set. The *co-lifting* of P is the poset $P^\top := P \cup \{\top\}$, where $\top \notin P$, with the order:

$$x \leq \top \quad \text{for all } x \in P^\top, \quad x \leq_{P^\top} y \iff x \leq_P y \quad \text{for all } x, y \in P.$$

Thus $P^\top$ is P with a new **top element** $\top$ adjoined.

0.1.9 Consistency

Definition 0.1.26 (Consistency). Let $(S, \leq)$ be a partially ordered set. Elements $x, y \in S$ are *consistent* (or *compatible*) if they admit a common upper bound:

$$\exists z \in S, \quad x \leq z \wedge y \leq z.$$

0.2 Ordered Sets

0.2.1 Down-Sets and Up-Sets

Definition 0.2.1 (Down-Set). Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is a *down-set* (or *decreasing set*, or *order ideal*) if it is closed downward:

$$\forall x \in Q, \forall y \in P, \quad y \leq x \implies y \in Q.$$

Definition 0.2.2 (Up-Set). Let $(P, \leq)$ be a partially ordered set. A subset $Q \subseteq P$ is an *up-set* (or *increasing set*, or *order filter*) if it is closed upward:

$$\forall x \in Q, \forall y \in P, \quad x \leq y \implies y \in Q.$$

0.2.2 Families of Sets

Proposition 0.2.3 (Predicates and Power Sets Are Order-Isomorphic). *The map $\Phi : (\mathcal{P}^*(X), \leq) \rightarrow (\mathcal{P}(X), \subseteq)$ is an *order-isomorphism*. Go to proof.*

Proofs

Capstone